C3 Problems: differential calculus I

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No chain rule is needed anywhere in this set. "Bereken exact" means no decimal approximations; "bereken" means a GR or calculator may be used.

Practice

1. Given $f(x) = x^3 - 2x$.

a. Compute the average rate of change of f on $[1, 3]$.
b. Simplify the difference quotient of f on $[2, 2+h]$ for $h \neq 0$, and use it to find $f'(2)$.

2. Find the derivative from the definition $f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$.

a. $f(x) = x^2 - 3x$
b. $f(x) = \dfrac{1}{x+1}$

3. Differentiate. Use the product or quotient rule where it helps, and simplify.

a. $f(x) = 4x^5 - 3x^2 + 7$
b. $f(x) = 6\sqrt{x}$
c. $f(x) = \dfrac{2}{x^3}$
d. $f(x) = (x^2+1)(x^3 - x)$: once with the product rule and once by expanding first
e. $f(x) = \dfrac{3x - 1}{x + 2}$
f. $f(x) = x^2\sqrt{x}$

4. Find the equation of the tangent line to the graph of $f(x) = x^3 - 3x$ at the point with $x = 2$.

5. Given $f(x) = 2x^3 + 3x^2 - 12x$. Bereken exact the extreme values of f and say which is a maximum and which a minimum.

6. Given $f(x) = x^4 - 8x^2$.

a. On which intervals is f increasing, and on which decreasing?
b. Give all extreme values.

7. Given $f(x) = x^4 - 6x^2 + 1$. Bereken exact the coordinates of the inflection points.

8. A particle moves along a line with position $s(t) = 2t^3 - 15t^2 + 36t$ (metres, t in seconds, $t \geq 0$).

a. Find the velocity $v(t)$ and acceleration $a(t)$.
b. At which moments is the particle at rest?
c. At which moment is the acceleration zero? What is the velocity then?

9. Find the equation of the tangent line to the graph of $f(x) = x^2 + 1$ that is parallel to the line $y = 4x + 7$.

CE-level

10. From a square sheet of cardboard of 30 by 30 cm, equal squares of side x cm are cut from the four corners. The sides are folded up to make an open box.

a. Toon aan that the volume is $V(x) = x(30-2x)^2$, and give the domain.
b. Bereken exact for which x the volume is maximal, and give the maximum volume.

11. Given $f(x) = \dfrac{x^2}{x-1}$.

a. Toon aan that $f'(x) = \dfrac{x^2-2x}{(x-1)^2}$.
b. Bereken exact the extreme values of f.

12. For every p the function $f_p(x) = x^3 - px$ is given.

a. For which values of p does f_p have two extreme values?
b. Toon aan that for all these p the points of the graph where f_p has an extreme value lie on the curve $y = -2x^3$.

13. Given $f(x) = x^2$ and the point $P(1,-3)$, which does not lie on the graph. Bereken exact the equations of the tangent lines to the graph of f that pass through P.

14. A farmer has 120 m of fencing. He fences a rectangular field along a straight river; the side along the river needs no fence. Calculate the dimensions that give the largest area, and that area.

15. Given $f(x) = \sqrt{x}$ and the point $A(4,0)$. P is a point on the graph of f. Bereken exact the smallest possible distance AP. *Hint:* minimise the square of the distance.

16. Given $f(x) = x^3 - 3x^2 + 4$.

a. Bereken exact the inflection point of the graph of f.
b. Give the equation of the tangent line at the inflection point (the *buigraaklijn*).
c. Toon aan that this tangent line has no other point in common with the graph.

17. Given $f(x) = \dfrac{1}{x}$ with $x > 0$. The tangent line at the point with $x = a$ intersects the x-axis at A and the y-axis at B. Toon aan that the area of triangle OAB is 2 for every $a > 0$.

18. A closed cylindrical can must contain 500 cm³.

a. Toon aan that the surface area is $S(r) = 2\pi r^2 + \dfrac{1000}{r}$, with r the radius in cm.
b. Calculate the radius that minimises S, in two decimals, and show that the height is then equal to the diameter.

19.

a. Show that $f(x) = x^4$ has $f''(0) = 0$ but that $(0,0)$ is not an inflection point.
b. Given $g(x) = x^5 - 5x^4$. Bereken exact all inflection points of the graph of g.

20. Of a function f it is known that $f'(x) = (x-1)^2(x+2)$. (The formula of f itself is not given.)

a. For which x does f have an extreme value? Maximum or minimum?
b. For which x does the graph of f have an inflection point?

Challenge

21.

a. Prove by induction on n, using only the product rule and $(x)' = 1$, that $(x^n)' = nx^{n-1}$ for all positive integers n.
b. Use the quotient rule to prove that $(x^{-n})' = -nx^{-n-1}$ for all positive integers n.

22.

a. Prove from the definition that $\left(\frac{1}{g}\right)' = -\frac{g'}{g^2}$ (where $g(x) \neq 0$ and g is differentiable).

b. Use (a) and the product rule to derive the quotient rule.

23. Prove that the graph of every cubic function $f(x) = ax^3 + bx^2 + cx + d$ ($a \neq 0$) is point-symmetric in its inflection point.

24. Given $f(x) = x^3$ and a number $a \neq 0$.

a. Find the tangent line at $x = a$.

b. Prove that this tangent line meets the graph again at exactly one other point, with $x = -2a$.

25. Find a and b such that $f(x) = x^3 + ax^2 + bx$ has a local maximum at $x = -1$ and an inflection point at $x = 1$. Verify that it really is a maximum.

26. Prove from the definition that $(\sqrt{x})' = \frac{1}{2\sqrt{x}}$ for $x > 0$. *Hint:* multiply numerator and denominator of the difference quotient by $\sqrt{x+h} + \sqrt{x}$.